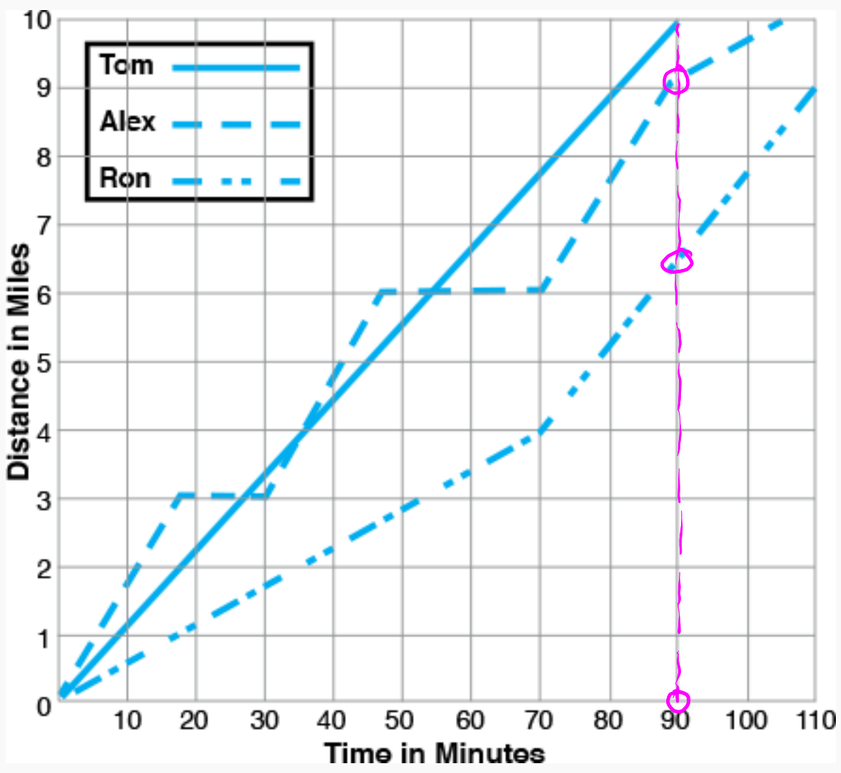
Comparing Functions

On a test, you may be asked to compare functions and draw conclusions. Sometimes, two or more functions may be drawn on the same graph.

Example 1

Hide

Tom, Alex, and Ron decide to run a 10-mile race. Data is collected at checkpoints along the course of the race. Using the data, each run can be expressed as a function and drawn on a graph. Which runner finished first?



Examine the three lines. Who ran 10 miles in the least number of minutes?
Tom finished the 10-mile run in 90 minutes. At 90 minutes, Alex had run about 9 miles and Ron between 6 and 7 miles. **Tom won the race.**

You may be asked to draw conclusions about functions that are shown in different ways. Make sure you understand how the presentations are related before you answer the questions.

Example 2

Hide

Elise wants to plant trees that will benefit the environment. She searches the Internet to find information about sugar maples and blue spruce trees. At a diameter of 8 inches, which type of tree is better at reducing CO_2 in the atmosphere?

Blue Spruce Facts

Annual Growth of Trunk Diameter: 0.5 inch

CO_2 Reduction

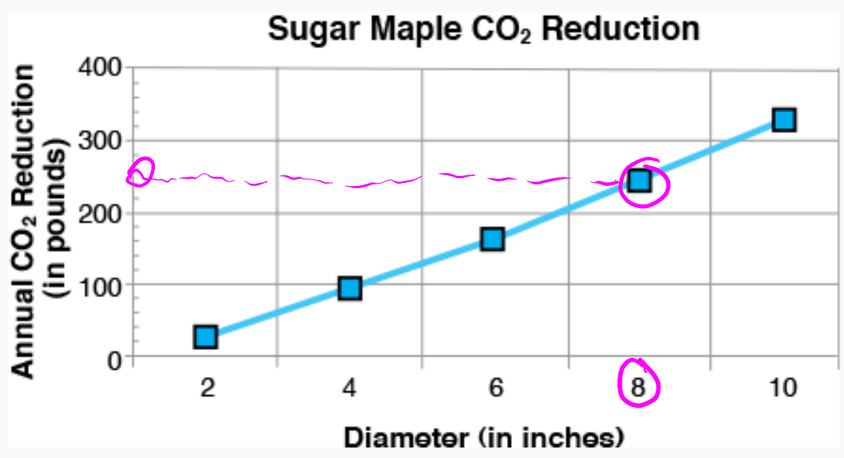
Diameter of Trunk	Annual CO_2 Reduction
2 in	28 lb
4 in	62 lb
6 in	106 lb
8 in	152 lb
10 in	203 lb

Annual Storm Water Recovery

Diameter of Trunk	Water Saved
2 in	123 gal
4 in	351 gal
6 in	723 gal
8 in	1,142 gal
10 in	1,626 gal

Sugar Maple Facts

The diameter of a sugar maple trunk grows at an average rate of 0.3 inch per year.
Its root system recovers storm water, which can be predicted using this function:
$g(x) = 4x^2 + 104x - 123$
where $g(x)$ represents gallons of water saved annually and x represents the diameter of the trunk in inches.



Read the first table. At 8 inches in diameter, a blue spruce removes 152 pounds of CO_2 from the atmosphere.
Read the line graph. At 8 inches in diameter, a sugar maple removes over 200 pounds of CO_2. **The sugar maple is better at reducing CO_2.**

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Lesson 5: Functions

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